Experiment 6

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

LO mapped: LO1

LO: Apply Agile methodologies and use Jira for Sprint Planning, task tracking and integrating with DevOps workflow.

Theory:

What is Jira?

Jira is an Agile project management and issue tracking tool developed by Atlassian. It helps software engineering teams plan sprints, track tasks, manage bugs, and monitor releases across customizable workflows.

Core Concepts & Hierarchy

- Project: A shared workspace representing a product, application, or team initiative.
- Issue Hierarchy:
  - Epic: A large objective or high-level feature broken down into smaller tasks (e.g., *Core Infrastructure*).
  - Story: An end-user feature written from a user perspective (e.g., *Storefront UI*).
  - Task / Bug: Technical implementations or defects to resolve.
  - Sub-task: Granular steps within a story or task.
- Product Backlog: A prioritized list of all pending features, enhancements, and bug fixes awaiting assignment.
- Sprint & Scrum Board: A time-boxed cycle (typically 1–2 weeks) where team members pull items from the backlog into active workflow columns (*To Do*, *In Progress*, *In Review*, *Done*).
- Story Points: Relative estimates assessing effort, technical complexity, and risk for each work item.

Jira Setup & Deployment Models

- Jira Cloud (SaaS): Hosted directly by Atlassian. No manual server maintenance or database installations required; teams create an organization site (e.g., company.atlassian.net) and get instant access via a web browser.
- Jira Data Center (Self-Managed): Installed on private infrastructure or dedicated cloud servers (AWS/Azure) for enterprises requiring localized data governance, custom security policies, and high-availability clustering.

DevOps Integration & Automation Lifecycle

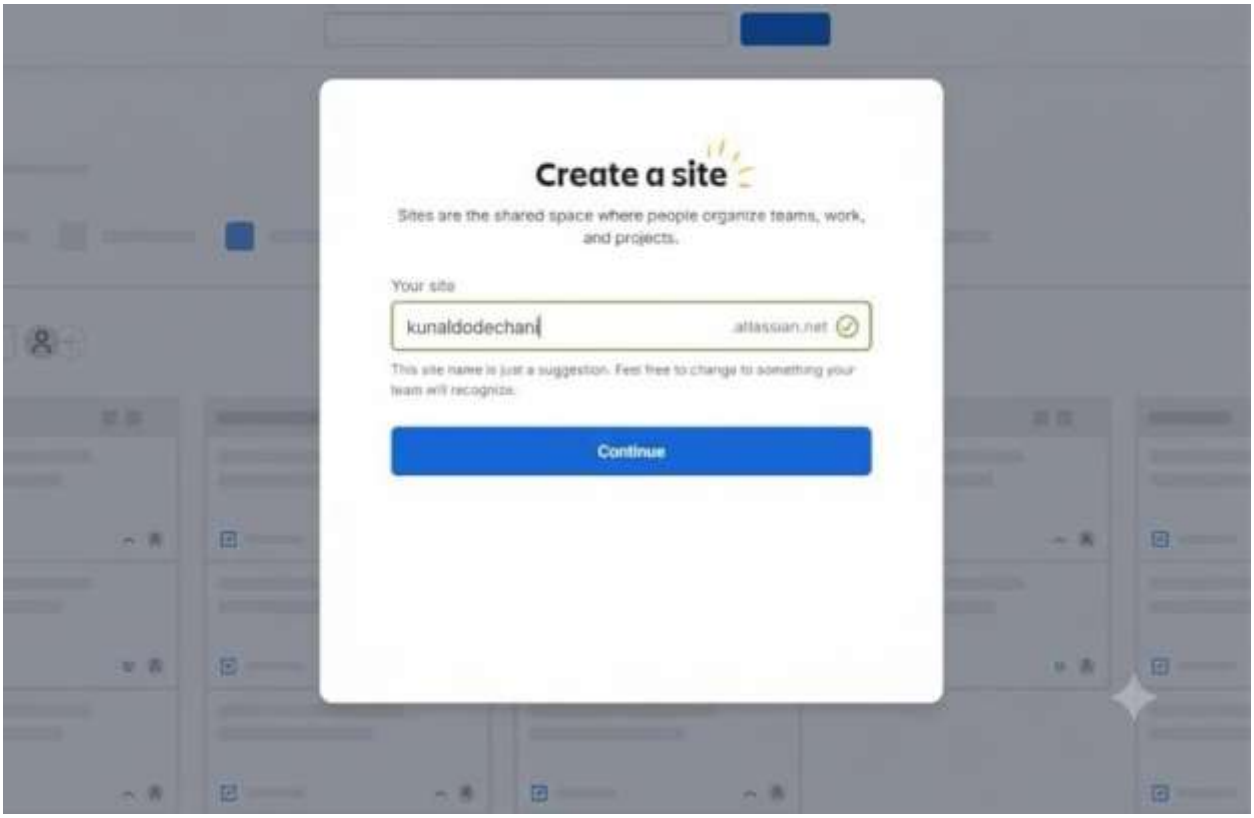
- Version Control Linkage: Integrates with code repositories (such as GitHub, GitLab, or Bitbucket) through official marketplace apps (e.g., *GitHub for Jira*).
- Smart Issue Keys: Mentioning ticket identifiers (e.g., SCRUM-4, SCRUM-6) in branch names, commit messages, or Pull Request titles binds VCS activity directly to the Jira issue panel.

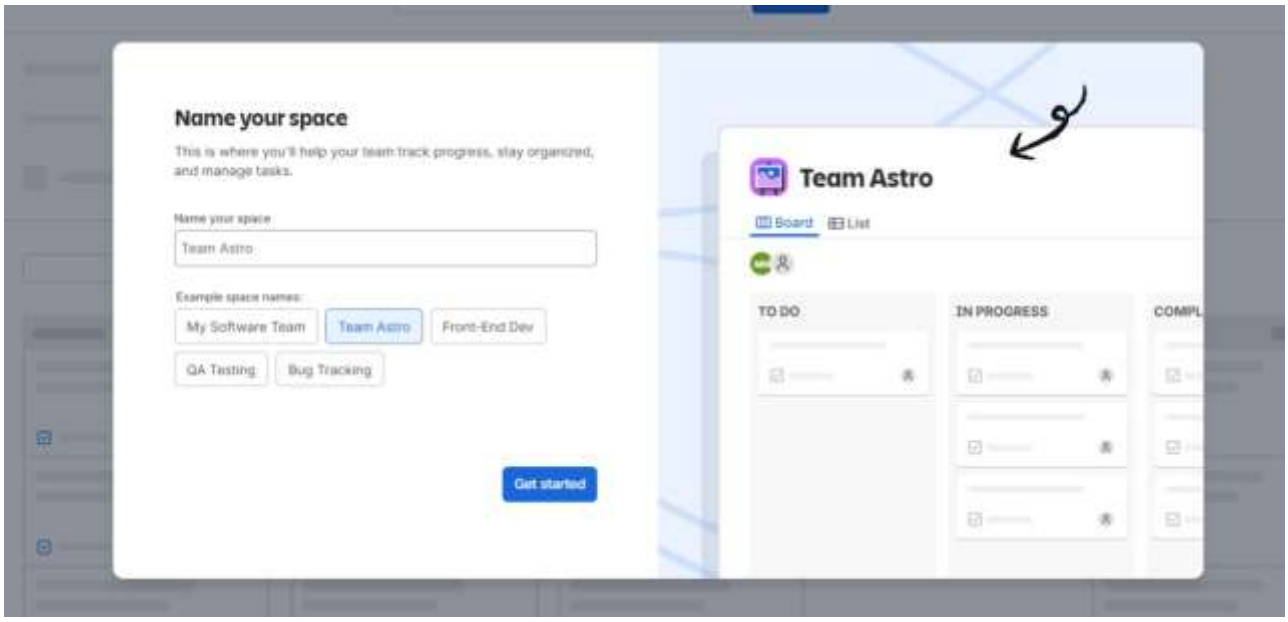
- Jira Automation Rules: Event-driven rules that trigger state transitions across the board (e.g., automatically shifting a ticket to *In Review* when a Pull Request is opened, or transitioning to *Done* upon a merged PR).

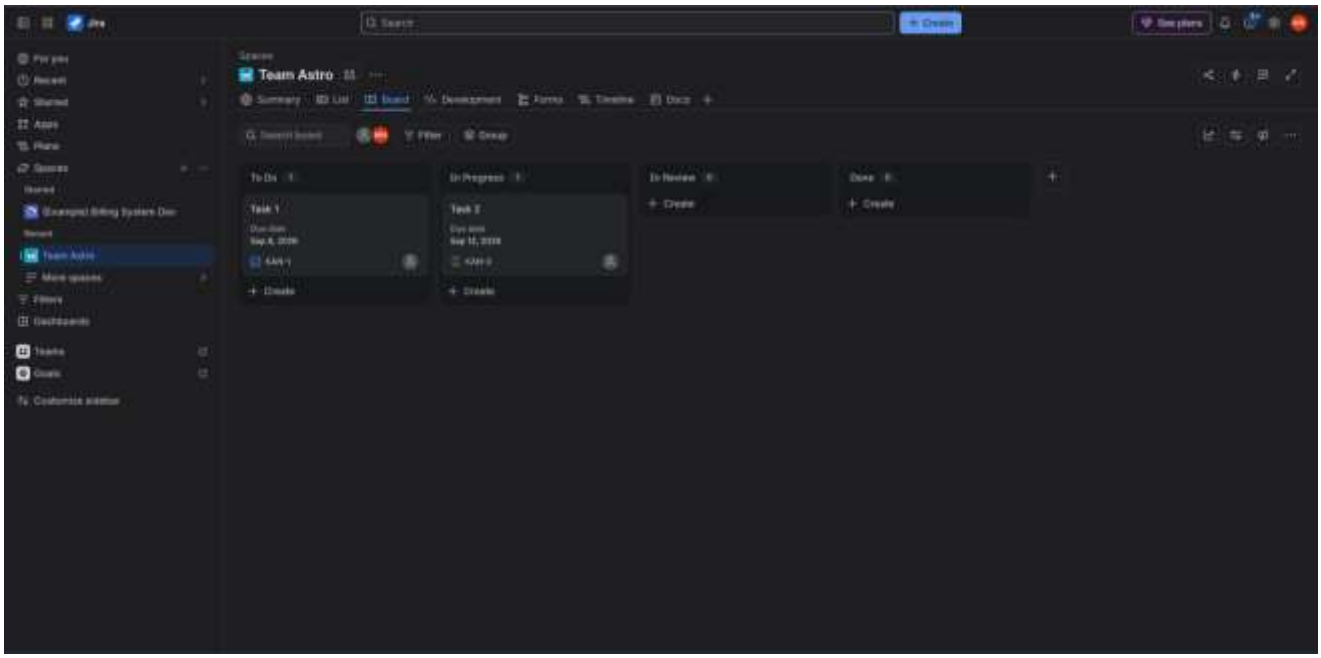
Team Collaboration & Access Management

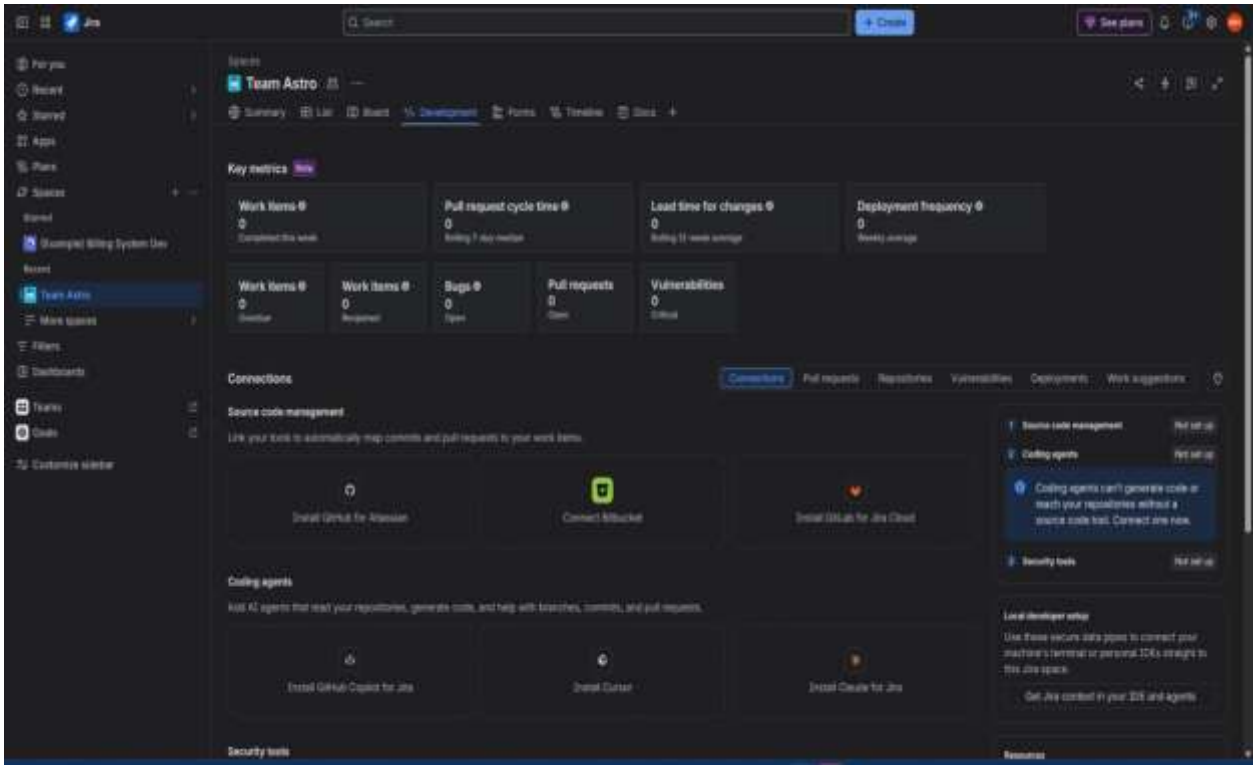
- Role-Based Access Control (RBAC): Access is governed at the organization and project levels via roles (*Administrator*, *Member*, *Viewer*).
- Work Allocation: Tickets are assigned directly to individual developers, allowing visibility into sprint capacity, bottlenecks, and velocity reports.

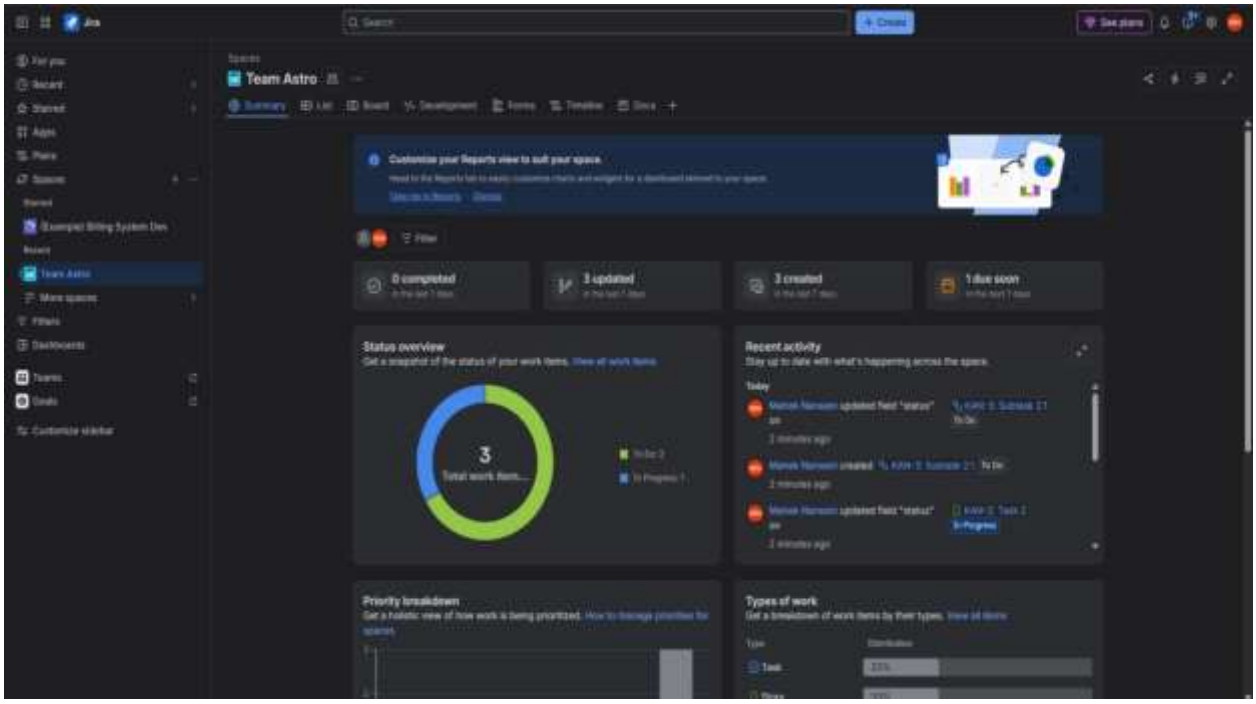
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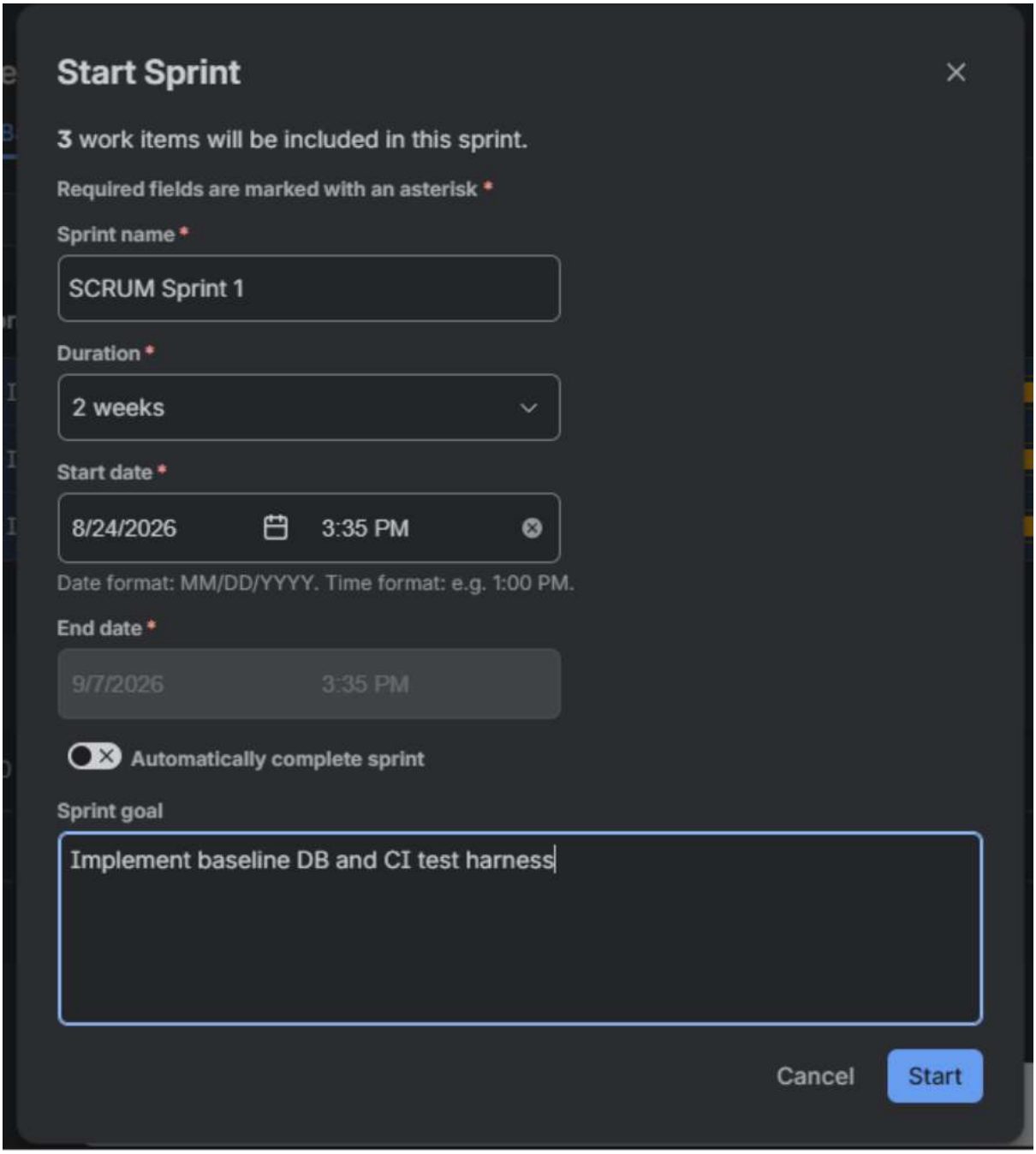


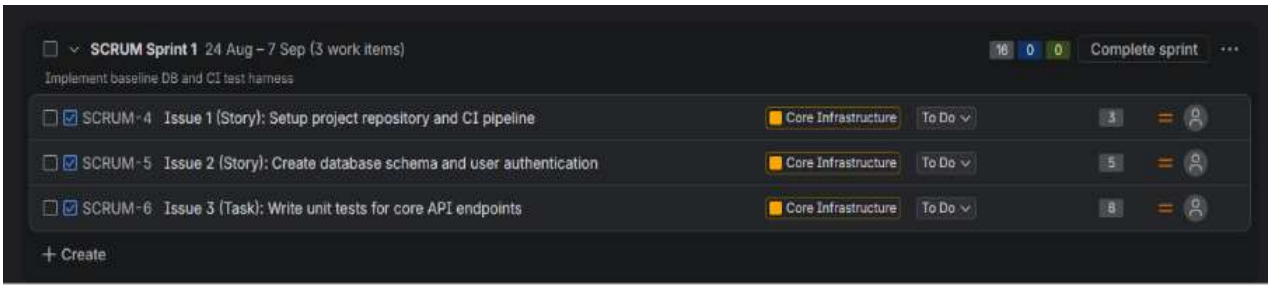

Name your space
This is where you'll help your team track progress, stay organized, and manage tasks.
Name your space
Team Astro
Example space names:
My Software Team
Team Astro
Front-End Dev
QA Testing
Bug Tracking
Get started
Team Astro
Board
List
TO DO
IN PROGRESS

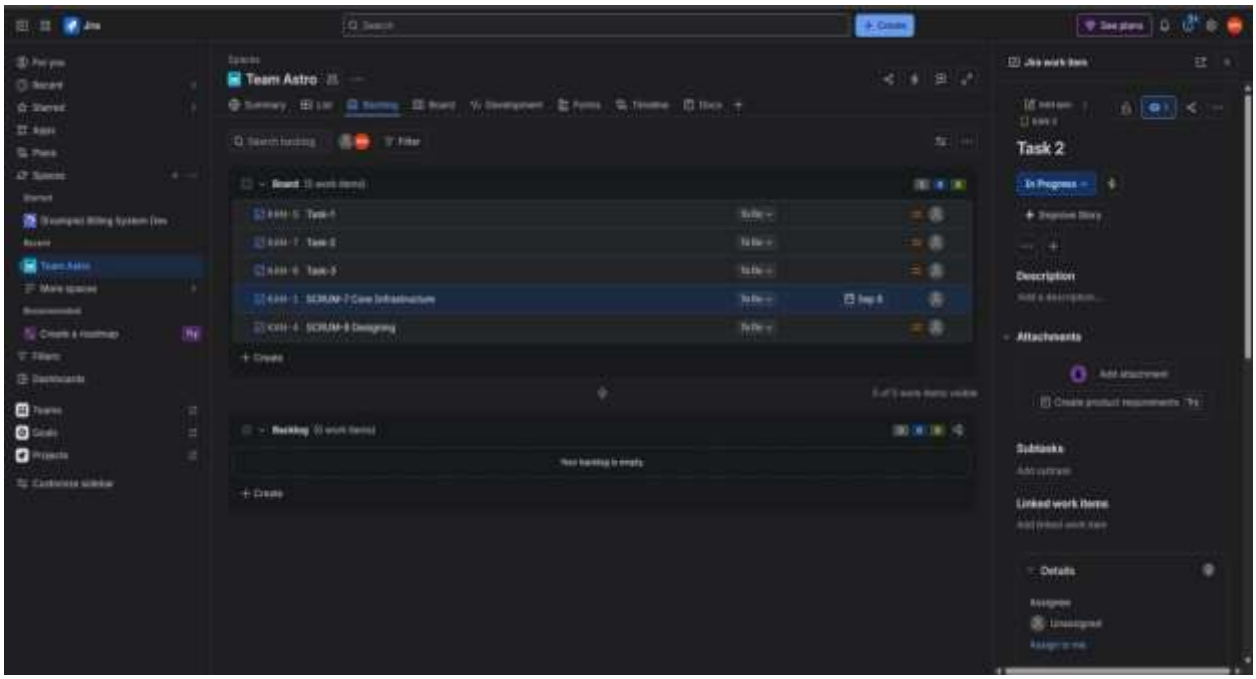

Team Astro

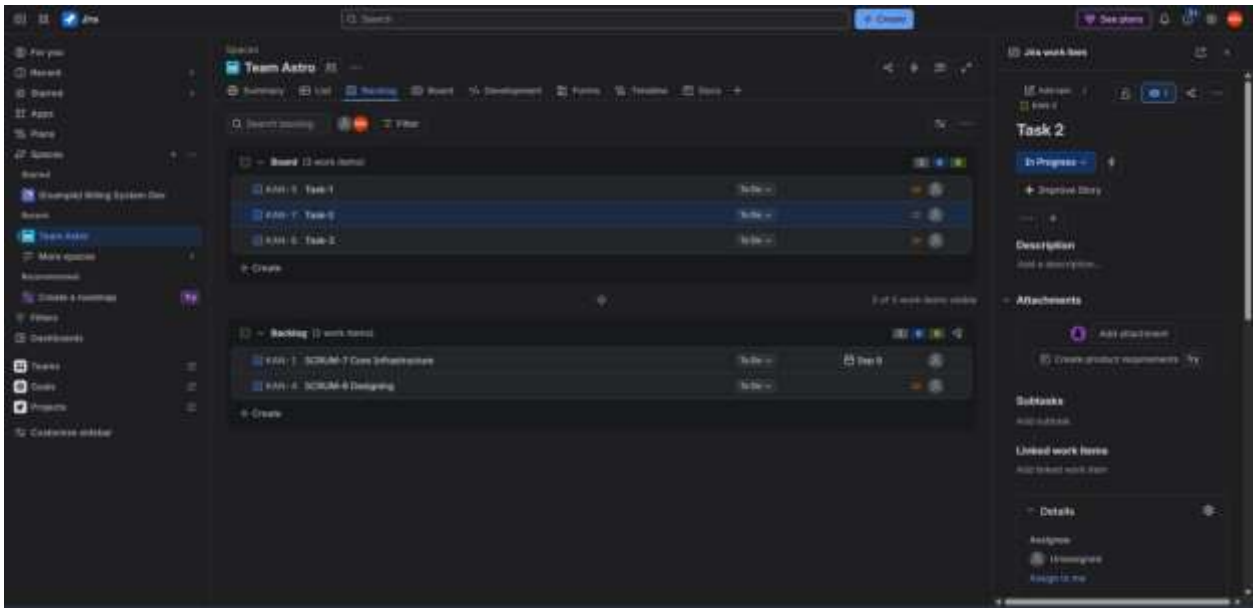

Team Astro
Key metrics
Work items
Pull request cycle time
Lead time for changes
Deployment frequency
Bugs
Pull requests
Vulnerabilities
Connections
Source code management
Coding agents
Security tools


Team Astro
Customize your Reports view to suit your space
0 completed
3 updated
3 created
1 due soon
Status overview
Recent activity
Priority breakdown
Types of work


Start Sprint
3 work items will be included in this sprint.
Required fields are marked with an asterisk *
Sprint name *
SCRUM Sprint 1
Duration *
2 weeks
Start date *
8/24/2026
3:35 PM
Date format: MM/DD/YYYY. Time format: e.g. 1:00 PM.
End date *
9/7/2026
3:35 PM
Automatically complete sprint
Sprint goal
Implement baseline DB and CI test harness
Cancel
Start


SCRUM Sprint 1 24 Aug – 7 Sep (3 work items)
Implement baseline DB and CI test harness
Complete sprint
SCRUM-4 Issue 1 (Story): Setup project repository and CI pipeline
Core Infrastructure
To Do
SCRUM-5 Issue 2 (Story): Create database schema and user authentication
Core Infrastructure
To Do
SCRUM-6 Issue 3 (Task): Write unit tests for core API endpoints
Core Infrastructure
To Do
Create





Conclusion

Integrating Jira with GitHub established an automated, end-to-end Agile development lifecycle for our E-commerce application. We learned how to organize backlogs into active sprints, assign tasks across team members, and track real-time progress using Scrum boards. By referencing issue keys in branch names and Pull Requests, we enabled seamless traceability between our codebase and Jira tickets. Furthermore, setting up GitHub Actions CI pipelines and Jira Automation demonstrated how modern DevOps workflows eliminate manual overhead by automating testing and board transitions upon code merges.